How AI Learns and Solves Problems

SMaiLE Project

Key Information

Target Group: 8 - 12 y.o.

Duration: Approx. 3–4 hours (modular)

Key Learning Goals:

1. **AI Concepts:** Understand how AI learns through Supervised Learning, Large Language Models, and Computer Vision.
2. **Real-World Application:** Match different AI types to real problems in sustainability and ecology.
3. **Critical Thinking:** Differentiate between human and AI-generated content.
4. **Design Thinking:** Prototype a conceptual AI solution for a community problem.

Learning Outcomes

Students will be able to:

KNOWLEDGE & UNDERSTANDING:

- Define Artificial Intelligence and describe in simple terms how it learns.
- Identify key AI learning paradigms such as supervised learning, Computer Vision, and Large Language Models.
- Understand that AI can be used to support innovation in fields like sustainability.

SKILLS & ABILITIES:

- Differentiate between human and AI-generated content.
- Work collaboratively to research and present ideas.
- Design a basic concept for an AI-powered solution.

ATTITUDES & VALUES:

- Develop critical thinking about AI capabilities and limitations.
- Understand the importance of safety, ethics, and responsibility when using AI.

European Dimension / Erasmus+ Connection

- **Bridging the Digital Gap:** Builds practical AI knowledge for students and teachers.
- **Teacher Development:** Can support hands-on professional development for educators.
- **Global Challenges:** Connects AI learning with sustainability and ecological problems.

1. Resources and Tools

- **Research Resources:** Computers, tablets, projector, screen, and recommended websites.
- **Digital Tools:** Video examples, Mentimeter or Kahoot, and optional Padlet for reflection.
- **Materials:** Matching cards or a digital matching game, whiteboard or flipchart, and markers.

Classroom Support Materials

- Prepare a worksheet, a simple task checklist, and a visible timer for group activities.
- Use introductory slides and Google Forms for reflection or quick assessment.

2. Working Methods

- **Inquiry Learning:** Encouraging questions and independent research.
- **Gamification:** Using “Find the AI Impostor” and an AI matching game.
- **Active Learning:** Group work, design prototyping, and peer evaluation.
- **Analysis and Discussion:** Reflecting critically on AI versus human responses.

Activity Overview

Phase	Duration	Description
Intro	30 min	“AI in Our Lives” discussion, introductory video, and a quick quiz with Mentimeter or Kahoot.
Research	90–120 min	Introduction to different AI paradigms, industry research, and “Find the AI Impostor” game.
Creative	50–60 min	AI Matching Game followed by designing a conceptual AI solution for a real problem.
Reflection	20 min	Padlet or written reflection and self-assessment rubric.

3. Introduction and Motivation

Goal: Build a shared understanding of what AI is.

- Start with a short video and discussion about where students encounter AI in everyday life.
- Ask what AI can do well and what concerns students may already have.
- Use a quick quiz to assess prior knowledge.

4. Research and Learning

Goal: Explore different ways AI learns and where it is used.

- Introduce supervised learning, Large Language Models, and Computer Vision in simplified language.
- In pairs, students research a field such as healthcare, environment, or smart cities.

- Students identify which AI paradigm is being used and explain why.
- Play "Find the AI Impostor" where students compare human and AI-generated answers.

5. Creative Application

Goal: Apply AI ideas to real-world problems.

- Begin with a matching game that pairs real-world problems with suitable AI solutions.
- In small groups, students choose a local or ecological problem and design a conceptual AI solution.
- Each group explains the problem, the proposed AI system, its learning method, and possible risks.

6. Reflection and Evaluation

Goal: Consolidate understanding and connect it to responsible AI use.

- Students complete a self-assessment rubric.
- Reflection questions can include: "What was the most surprising thing I learned about AI?" and "How can we use AI in a fair and safe way?"
- End with a short discussion of benefits, risks, and responsible choices.